PHILIPPINE SOCIETY OF
ANIMAL NUTRITIONISTS

# 39th ANNUAL CONVENTION

## Fueling Progress: Shaping Animal Nutrition for a Competitive and Sustainable Future

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# The Evolving Ethanol Process: From Corn to Corn Fermented Protein (CFP)

## How process fractionation turns a fuel co-product into a concentrated, versatile feed protein

*Budi Tangendjaja, PhD*
*Technical Consultant*
*USGBC SEA-Oceania*

# The Evolving Ethanol Process: From Corn to Corn Fermented Protein (CFP)

## How process fractionation turns a fuel co-product into a concentrated, versatile feed protein

U.S. GRAINS & BIOPRODUCTS COUNCIL

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# Ethanol plants are becoming biorefineries

Corn fermented protein (CFP) upgrades the feed value of the non-starch fraction.

1

## Process shift

Modern plants recover oil, fiber and protein at selected points instead of recombining every stream as DDGS.

Separation before or after fermentation changes yield, composition and capital requirements.

~50%

## Product shift

Commercial CFP commonly targets about 50% crude protein on an as-fed basis. The protein includes corn and fermentation-derived microbial biomass.

Actual specifications vary by technology and plant.

AA + ME

## Formulation shift

Value depends on digestible amino acids, metabolizable energy, phosphorus, fiber and consistency. Crude protein alone cannot establish a fair replacement price.

Sources: U.S. Grains Council (2023); POET (2026).

DDGS: Dried Distillers Grains with Solubles

# Three generations of corn ethanol co-products

Each step creates more differentiated feed ingredients and more process optionality.

## DDGS

**1. Conventional dry grind**

Corn is milled, fermented and distilled. Remaining solids and solubles are dried together as DDGS.

## DCO

**2. Oil recovery + fractionation**

Centrifugation and targeted separation recover distillers corn oil and alter the fiber-to-protein balance.

## CFP

**3. Protein-focused biorefinery**

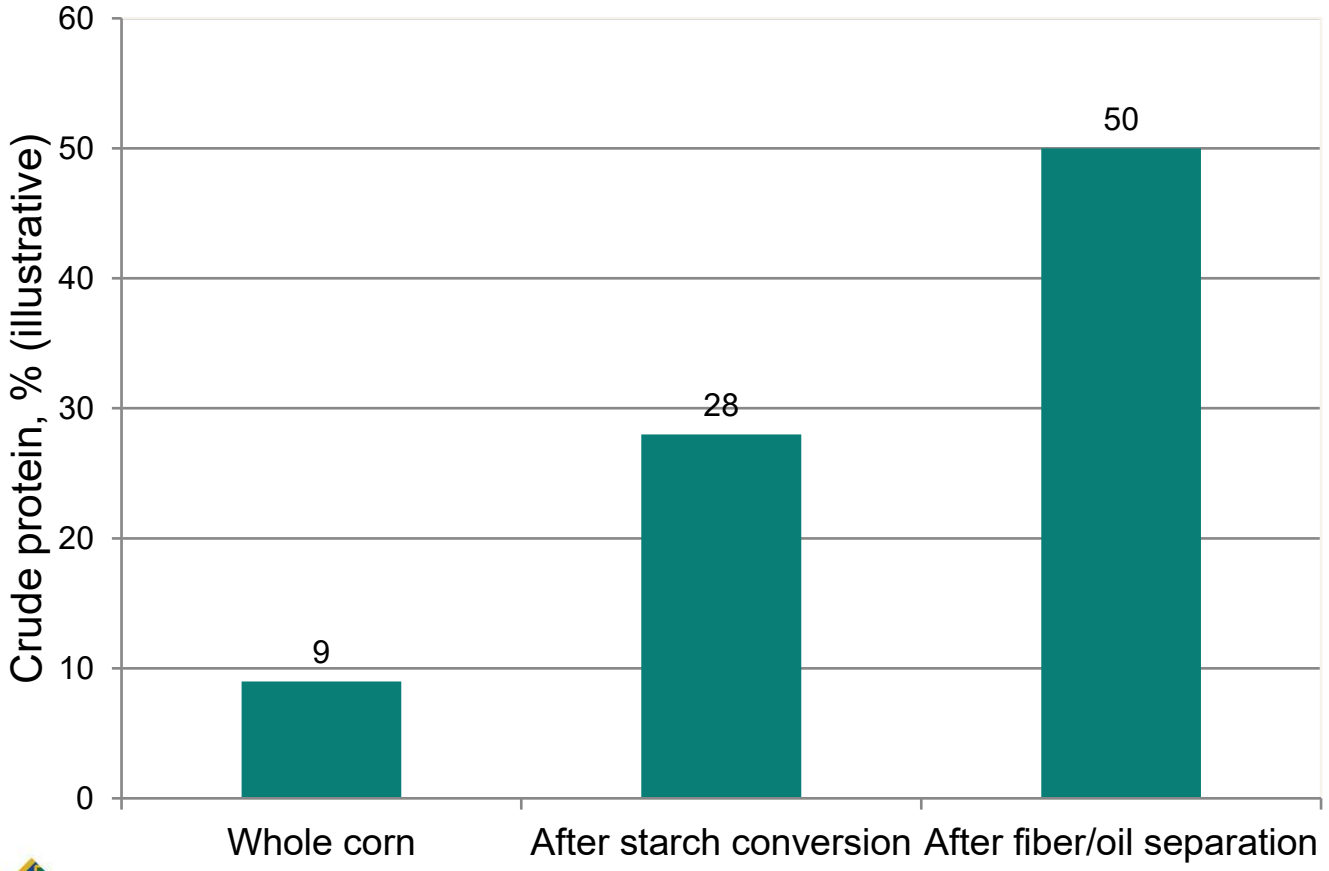
Front-end or back-end separation concentrates corn protein and yeast into CFP while producing lower-protein fiber streams.

Concepts synthesized from U.S. Grains Council High-Protein DDGS Handbook (2023).

DCO: Distillers Corn Oil

# The mass-balance logic starts with starch removal

Fermentation consumes starch. Separating fiber and oil raises protein concentration further.




Illustrative concentration pathway, not a universal plant mass balance. Sources: USGC (2023); Williams & Burton (2026).

## Concentration, not synthesis

Most of the increase comes from removing starch, oil and fiber mass from the protein-bearing solids.

## Fermentation adds microbial protein

Yeast biomass contributes amino acids and bioactive cell-wall fractions, although levels vary by process.

## Drying still matters

Excess heat can reduce reactive lysine and darken the product. Plant controls remain decisive.

# CFP can emerge from different separation routes

The technology route explains why products with the same protein guarantee can differ nutritionally.

| Route | Separation point | Primary mechanism | Typical implication |
|---|---|---|---|
| Front-end fractionation | Before fermentation | Mechanical or enzymatic removal of fiber, germ and/or protein | Cleaner starch feed to fermentation; distinct germ/fiber products |
| Back-end fractionation | After fermentation/distillation | Centrifugation, screening, washing and selective drying | Captures yeast-rich fine particles and concentrates protein |
| Hybrid process | Multiple points | Combines front-end preparation with post-fermentation recovery | Greater product control, with more equipment and operating complexity |

## Commercial naming is inconsistent

"High-protein DDGS," "Corn Protein Concentrate", "corn fermented protein," and branded names may describe products made by different routes. Formulators should buy against a complete specification and digestibility evidence.

Source: U.S. Grains Council High-Protein DDGS Handbook (2023).

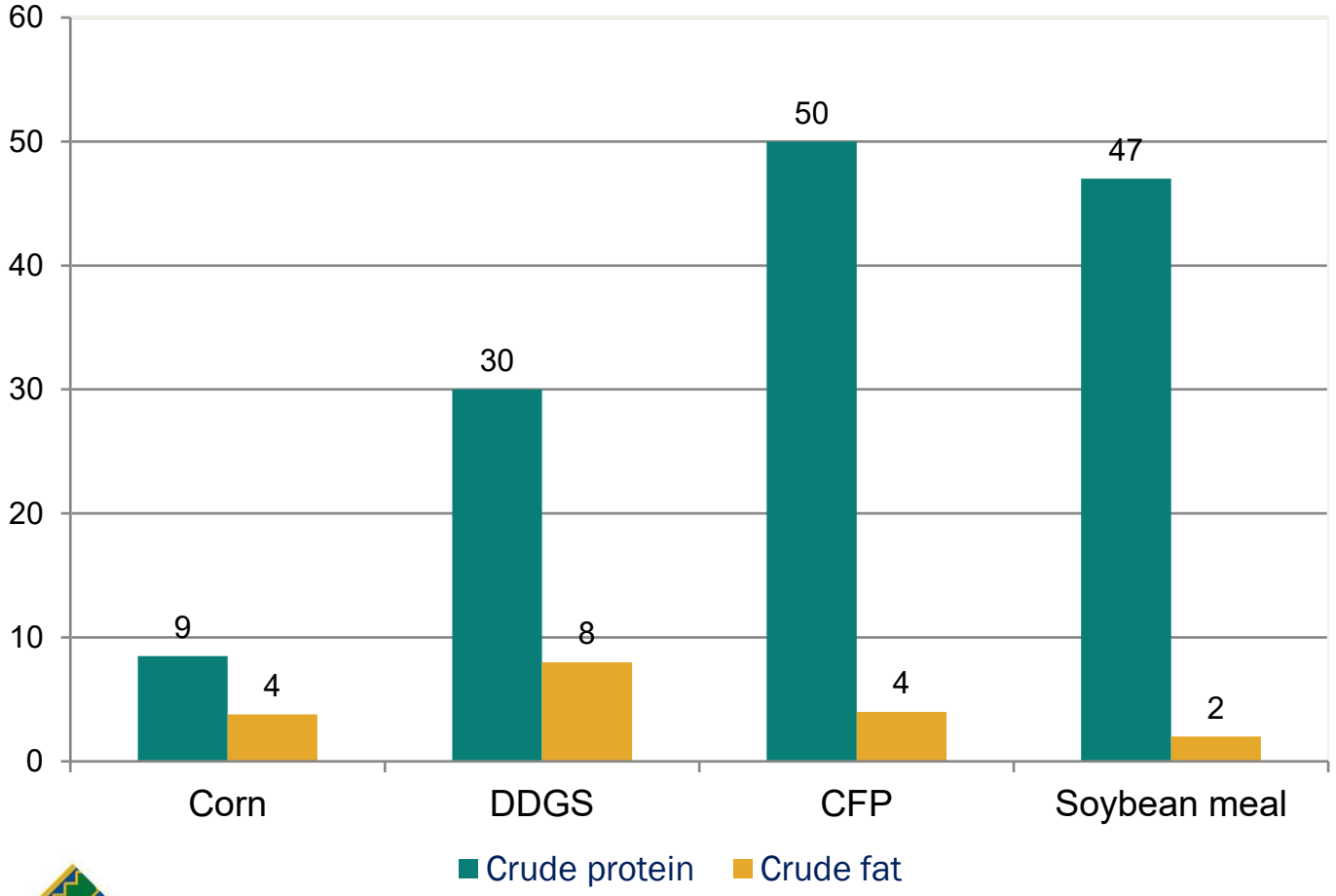
# Explanation of New Technology



- Why did the industry develop the technology?
  - Additional value/revenue streams
  - Precision nutrition
  - Capture additional ethanol or fat yield
  - Compete with higher-protein ingredients
  - Enter additional markets like aquaculture and pet food
- Basic premise – separate fractions of DDGS
- A lot of different technologies
  - Not necessarily better/worse – just different
  - Talk with supplier to get more details on process

# CFP fills the space between DDGS and concentrated protein meals

Representative values show positioning, not purchase specifications.




Representative ranges compiled from USGC (2023) and standard feed references; confirm supplier COA.

## Protein density

CFP reduces the gap between corn co-products and soybean meal in crude protein concentration.

## Different amino acid pattern

Corn protein contributes more leucine and less lysine than soybean meal. Yeast improves the profile but does not erase the difference.

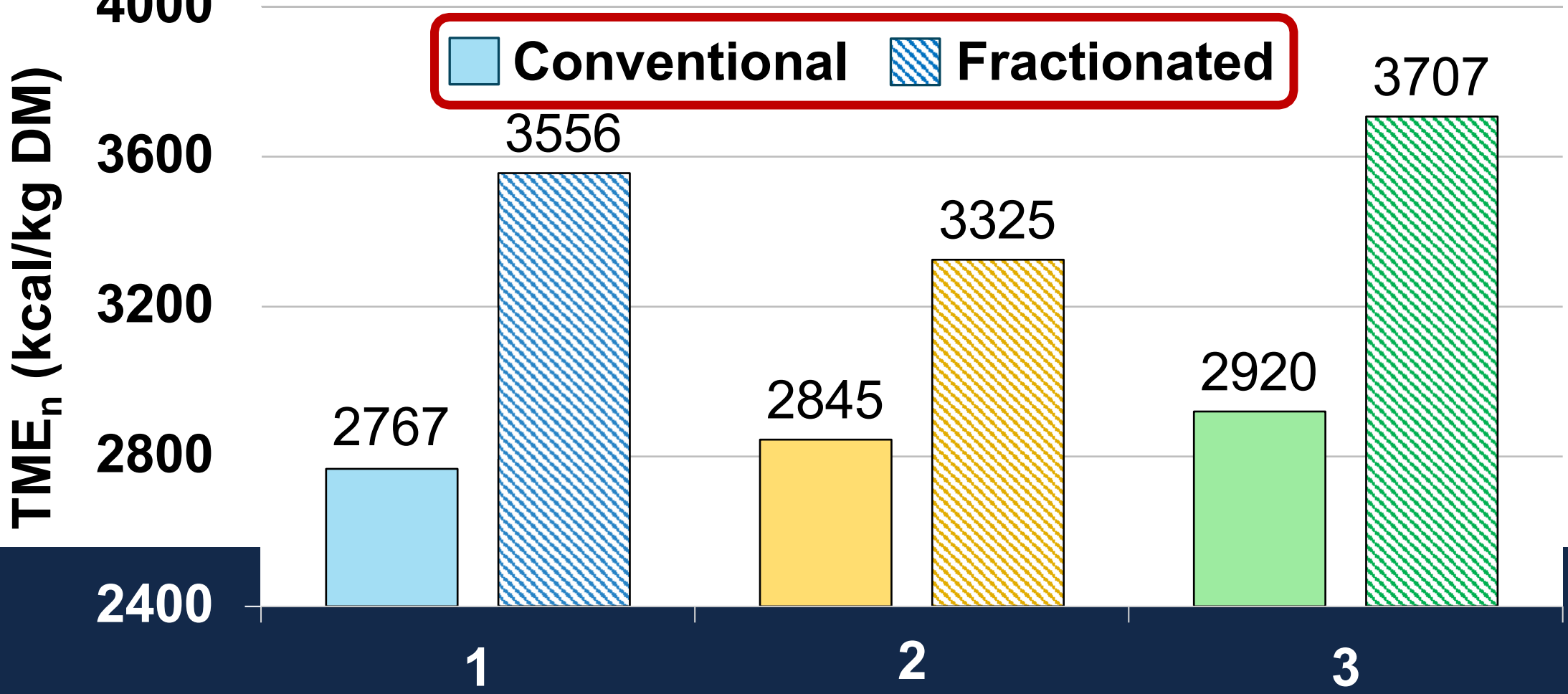
## Energy can remain valuable

Residual oil, digestible fiber and lower antinutritional factors can support competitive energy values.

# Range in nutritional composition of corn fermented protein sources for broilers (as-fed basis)

| Analyte | Range |
|---|---|
| Dry matter, % | 90.0 – 93.8 |
| Metabolizable energy, kcal/kg | 3,167 – 3,472 |
| Crude protein, % | 50.2 – 51.1 |
| Crude fat, % | 4.7 – 9.6 |
| Neutral detergent fiber, % | 22.4 – 27.5 |
| Acid detergent fiber, % | 4.5 – 20.0 |
| Ash, % | 2.2 – 5.1 |
| Ca, % | 0.01 – 0.05 |
| P, % | 0.68 – 1.02 |
| Na, % | 0.04 – 0.05 |

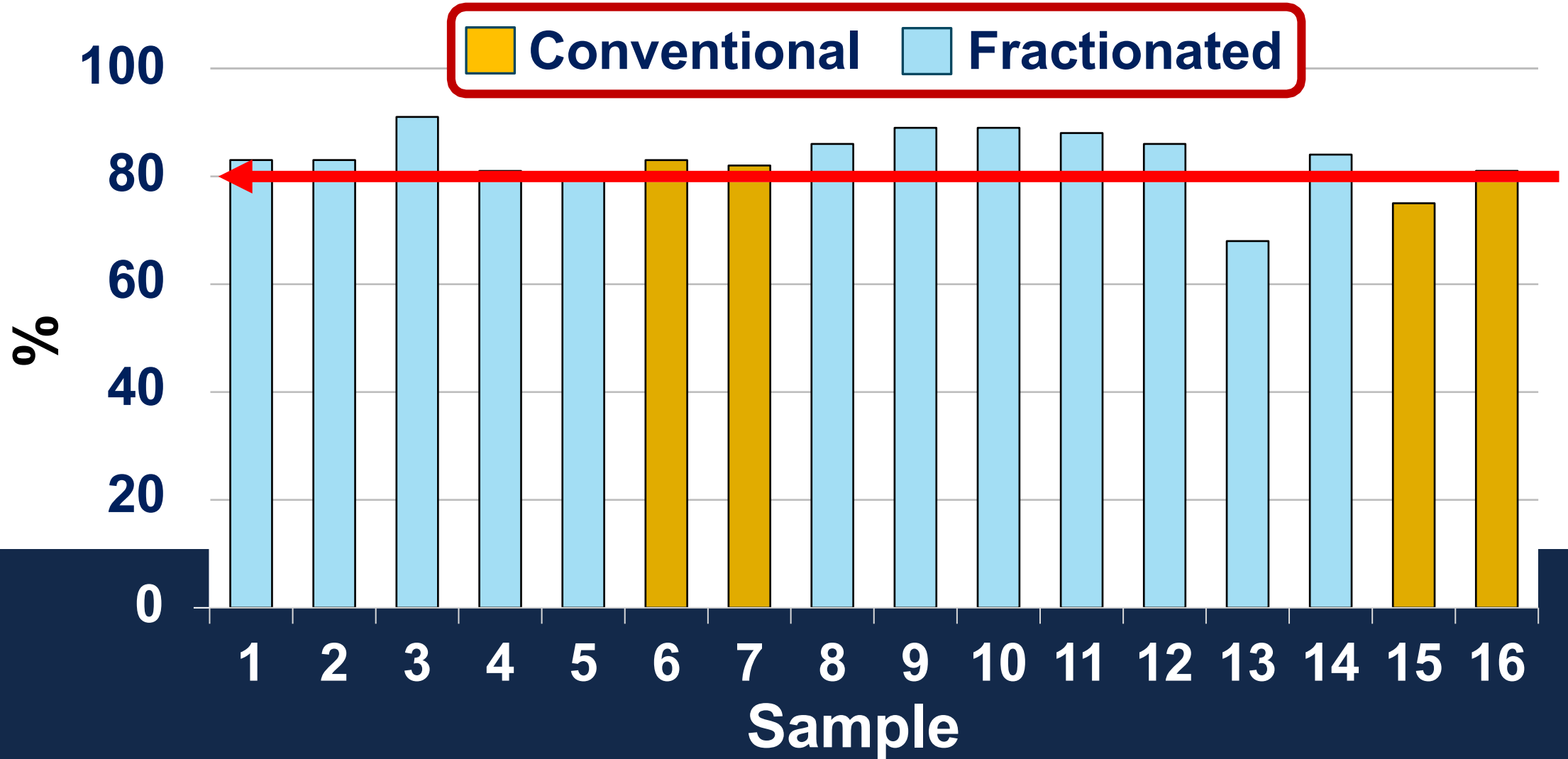
# Increased ME from Fractionation




ME: Metabolizable energy
TMEn: True Metabolizable Energy, N-corrected

Parsons et al. (2023)

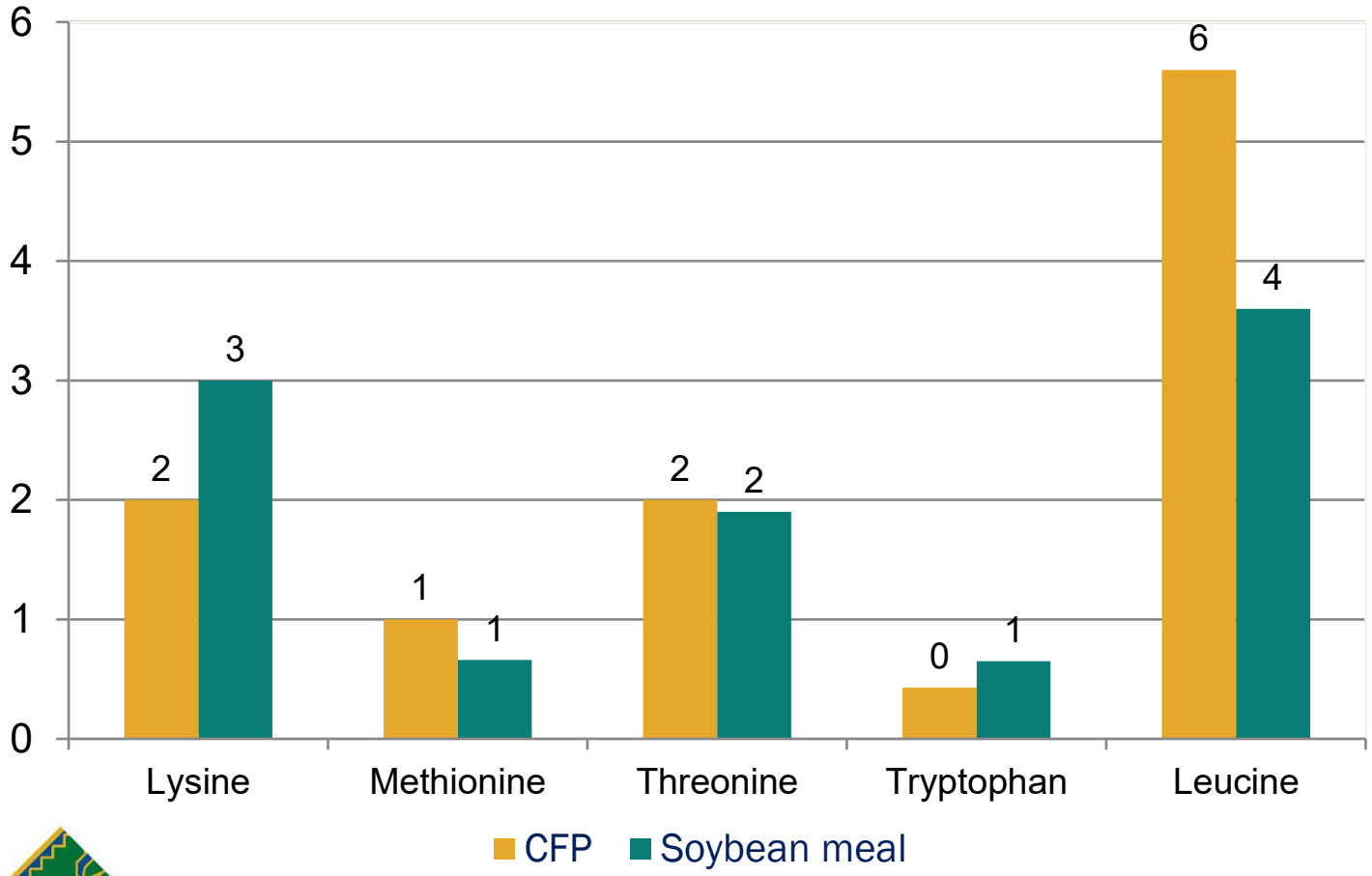
# Lysine Digestibility




Parsons et al. (2023)

# Protein percentage overstates value unless amino acids are digestible

CFP's practical value comes from SID amino acids delivered per unit of diet cost.




## Strength

Methionine and several branched-chain amino acids compare favorably on a concentration basis.

## Constraint

Lysine concentration remains below soybean meal and may show more heat sensitivity.

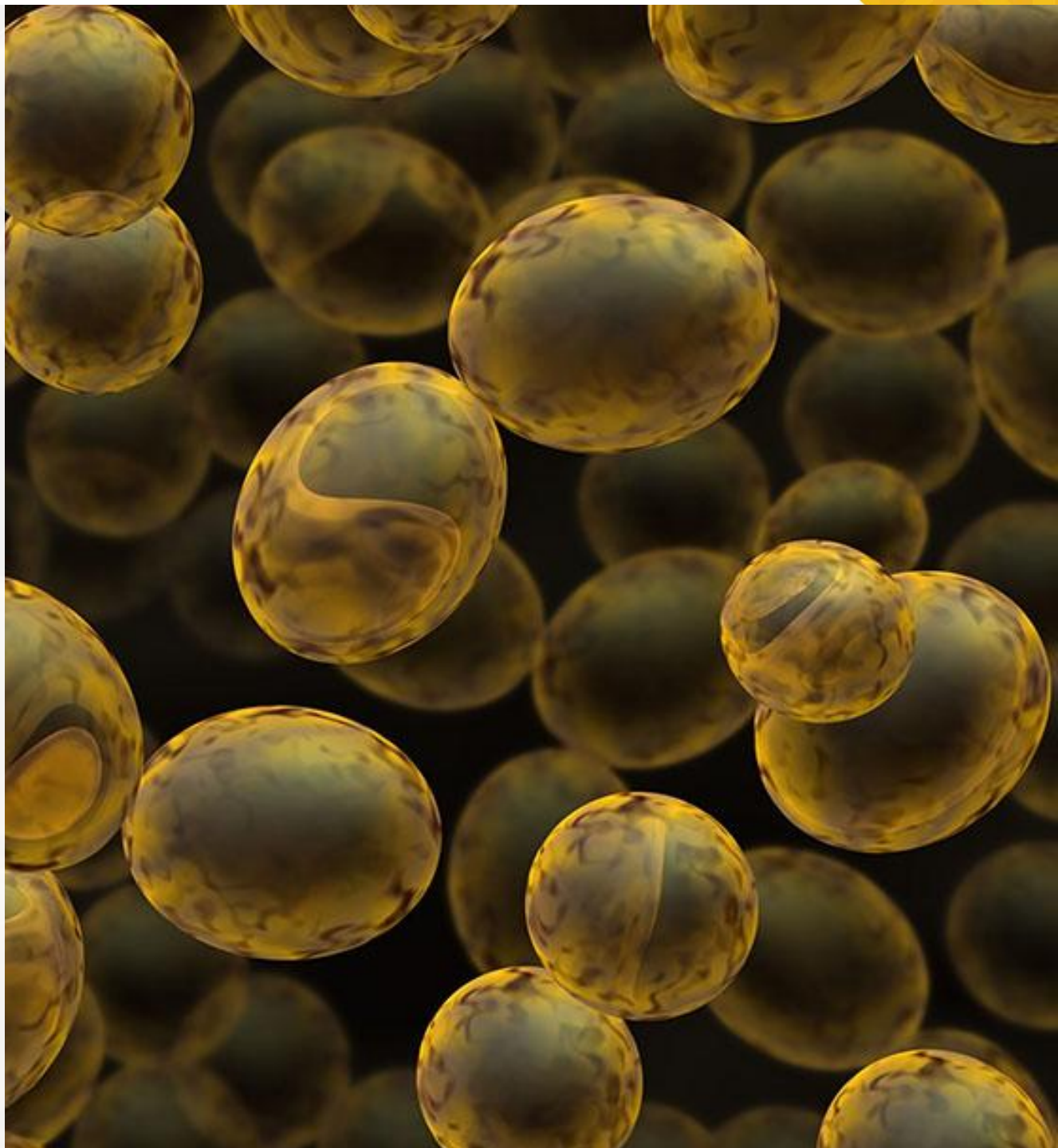
## Decision rule

Price digestible amino acids, not crude protein. Monitor BCAA

Illustrative as-fed amino acid concentrations; CFP example informed by public NexPro analysis. Confirm supplier matrix.

# Corn co-products contain various estimated amounts of spent yeast

- DDGS – 7-10%
- HP-DDG - ~16%
- Corn Fermented Protein – 20-29%



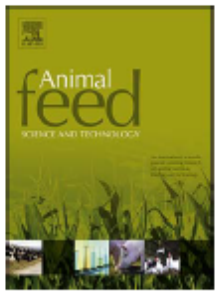
# Learn more about the potential role of residual yeast in corn co-products on animal health by reading this review article

Animal Feed Science and Technology 235 (2018) 60–76

Contents lists available at ScienceDirect

Animal Feed Science and Technology

journal homepage: www.elsevier.com/locate/anifeedsci




Review Article

## Yeast and yeast derivatives in feed additives and ingredients: Sources, characteristics, animal responses, and quantification methods

G.C. Shurson*

*Department of Animal Science, 385 ANSC/VM Bldg., 1988 Fitch Ave., University of Minnesota, St. Paul, MN 55108, USA*

# Poultry value depends on energy, SID amino acids and inclusion level

CFP can displace portions of soybean meal, DDGS and other proteins when its matrix is accurate.

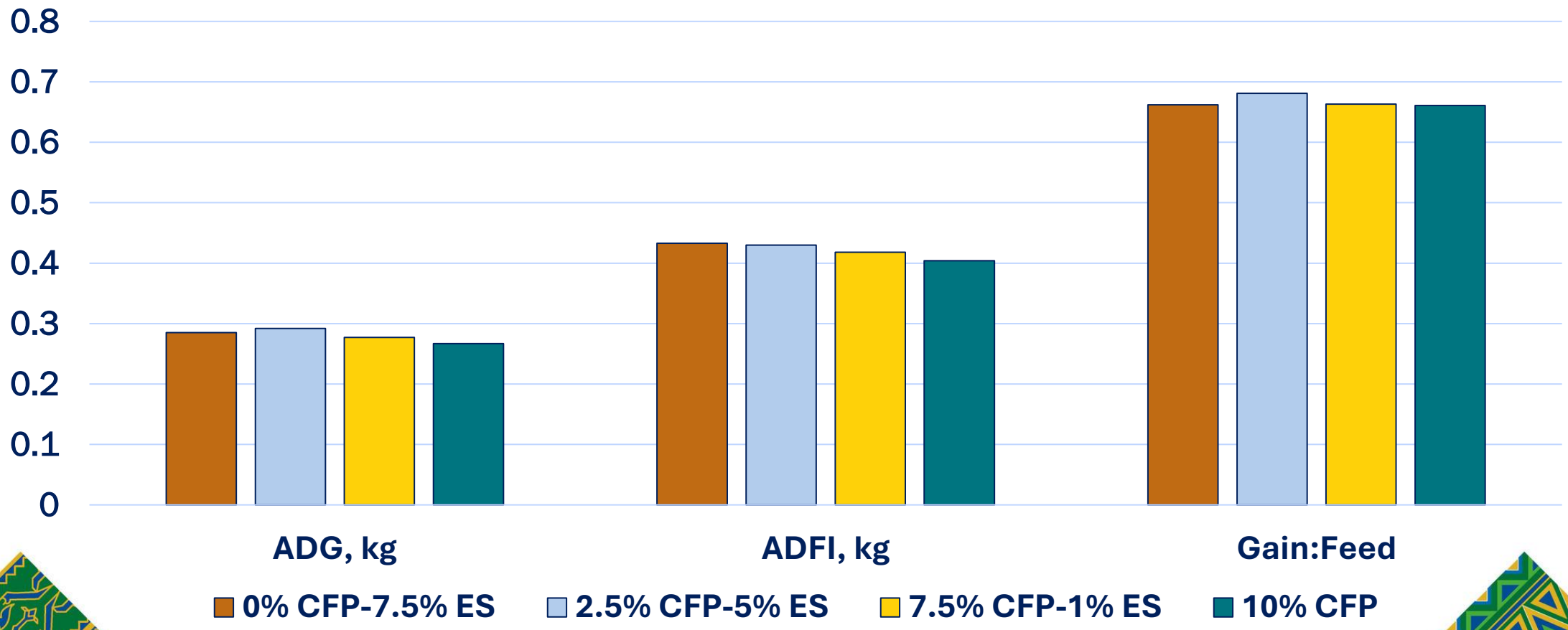
| Application | Potential Role | Formulation Watchpoint |
|---|---|---|
| Broiler starter | Concentrated protein with useful methionine and energy | Lysine balance, leucine excess, pellet quality |
| Broiler grower/finisher | Cost-effective partial protein replacement | Validate AMEn and SID coefficients by supplier |
| Layers | Protein source with corn pigments and low starch | Egg mass response, yolk color, phosphorus matrix |
| Turkey | Alternative protein in amino acid-balanced diets | High amino acid density increases matrix-error risk |

## Practical introduction protocol

Start with a conservative inclusion, reformulate on digestible nutrients, track feed intake and FCR, then expand only after performance and pellet-quality data agree with the matrix.

Source: USGC, Feeding Applications of CFP Co-Products in Poultry Diets (2023).

# Effects of feeding increasing amounts of corn fermented protein (CFP) to partially replace enzyme treated soybean meal (ES) in diets fed to weaned pigs on growth performance (day 8 to 21 post-weaning)




[a,b,c] Means with uncommon superscripts are different (P < 0.05).
Acosta et al. (2021)

# Summary of Swine Feeding Trial Results

Up to 16% corn fermented protein in Phase 1 and Phase 2 starter diets

Amino acid imbalance in corn fermented protein requires addition of synthetic lysine, threonine, and tryptophan

(and possibly valine and isoleucine)

to diets

# Aquaculture and pet food offer higher-value channels with higher proof requirements

Protein concentration helps, but functionality, palatability and physical quality become decisive.

| Market | Attraction | Barrier to Adoption |
|---|---|---|
| Aquaculture | Concentrated plant and microbial protein; low starch; potential pellet-binding benefits | Species-specific digestibility, palatability, fiber and amino acid balance |
| Pet food | Non-soy positioning, yeast story and protein density | Ingredient definition, color, flavor, stool quality and regulatory acceptance |
| Specialty young-animal feeds | Protein concentration and possible functional yeast fractions | Evidence threshold, microbial quality and batch consistency |

## Commercial upside rises with specification discipline

Premium channels require tighter control of odor, color, particle size, contaminants and traceability than commodity livestock feed.

Sources: POET (2026); USGC High-Protein DDGS Handbook (2023).

# Formula broiler starter using CFP and Shadow Price

Ingredients for MKT5 / P1 — Broiler Starter

| Code | Name | Solution Amount % | Price | * | Low Cost | High Cost | Previous Solution |
|---|---|---|---|---|---|---|---|
| 9 | Corn, U.S. | 57.595 | 320 | | 279 | 344 | 61.619 |
| 220 | SBM US dehulled | 20.956 | 480 | | 467 | 482 | 16.334 |
| 244 | US DDGS reduce fat | ↑ 10.000 | 340 | | 340 | 372 | 4.315 |
| 55 | MBM | ↑ 5.000 | 620 | | 620 | 743 | 5.000 |
| 41 | Corn Gluten Meal 62 | 2.568 | 708 | | 581 | 729 | |
| 65 | Crude Palm Oil | 1.418 | 1,050 | | 991 | 1,058 | 0.192 |
| 70 | Limestone | 0.728 | 35 | | | 285 | 0.721 |
| 75 | MDCPhosphate | 0.458 | 1,027 | | 263 | 1,099 | 0.580 |
| 94 | L-lysine HCl | ↑ 0.400 | 1,600 | | 1,600 | 2,170 | 0.400 |
| 92 | D,L-methionine | 0.270 | 6,000 | | 889 | 7,287 | 0.216 |
| 80 | Salt | 0.225 | 142 | | | 644 | 0.215 |
| 81 | Sodium Bicarb | 0.161 | 350 | | | 1,385 | 0.193 |
| 96 | L-threonine | 0.078 | 1,800 | | 598 | 3,184 | 0.062 |
| 100 | Choline Chloride 50% | ↓ 0.050 | 1,239 | | | 1,239 | 0.050 |
| 125 | Mineral Mix HC | 0.040 | 2,124 | | 2,124 | 2,124 | 0.040 |
| 120 | Vitamin Mix HC | 0.025 | 10,619 | | 10,619 | 10,619 | 0.025 |
| 101 | Choline Chloride 60% | 0.024 | 1,504 | | 1,057 | 4,781 | 0.032 |
| 97 | L-tryptophan | 0.004 | 7,080 | | 713 | 11,646 | 0.008 |

Cost $418/MT or PhP 26.2/kg

Ingredients for MKT5 / P1 — Broiler Starter

| Code | Name | Solution Amount % | Price | * | Low Cost | High Cost |
|---|---|---|---|---|---|---|
| 9 | Corn, U.S. | 61.619 | 320 | | 279 | 341 |
| 220 | SBM US dehulled | 16.334 | 480 | | 404 | 480 |
| 159 | Corn Fermented Protein | ↑ 10.000 | 500 | | 500 | 583 |
| 55 | MBM | ↑ 5.000 | 620 | | 620 | 705 |
| 244 | US DDGS reduce fat | 4.315 | 340 | | 335 | 372 |
| 70 | Limestone | 0.721 | 35 | | | 289 |
| 75 | MDCPhosphate | 0.580 | 1,027 | | 362 | 1,100 |
| 94 | L-lysine HCl | ↑ 0.400 | 1,600 | | 1,600 | 5,595 |
| 92 | D,L-methionine | 0.216 | 6,000 | | 796 | 10,253 |
| 80 | Salt | 0.215 | 142 | | | 289 |
| 81 | Sodium Bicarb | 0.193 | 350 | | 221 | 3,030 |
| 65 | Crude Palm Oil | 0.192 | 1,050 | | 890 | 1,051 |
| 96 | L-threonine | 0.062 | 1,800 | | 391 | 1,986 |
| 100 | Choline Chloride 50% | ↓ 0.050 | 1,239 | | | 1,239 |
| 125 | Mineral Mix HC | 0.040 | 2,124 | | 2,124 | 2,124 |
| 101 | Choline Chloride 60% | 0.032 | 1,504 | | 1,443 | 6,952 |
| 120 | Vitamin Mix HC | 0.025 | 10,619 | | 10,619 | 10,619 |
| 97 | L-tryptophan | 0.008 | 7,080 | | 5,962 | 29,970 |

Cost $406/MT or PhP 25.5/kg

KEY CONCLUSION

# CFP converts process precision into feed value

The opportunity is real when three conditions hold:

| | | |
|---|---|---|
| 1 | **A defined product** | New Development in Ethanol Process, Plant-specific product |
| 2 | **A defensible matrix** | Higher Metabolizable Energy, Digestible amino acids and Phosphorus |
| 3 | **A disciplined rollout** | Economic viability and animal-performance validation |

Questions ?

# The Evolving Ethanol Process: From Corn to Corn Fermented Protein (CFP)

## How process fractionation turns a fuel co-product into a concentrated, versatile feed protein

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